Cesar Briones

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Education

University of South Florida – **B.S. in Mechanical Engineering** Expected Dec 2026
Key Courses: Aerodynamics, Propulsion, Vibrations, Finite Element Methods GPA: 3.72/4.0

Skills

CAD/Coding:	Siemens NX, SolidWorks, Python, MATLAB, Simulink, C#, LaTeX, Epsilon3
Design:	GD&T, DFMA, Prototyping, CNC Milling Programming and Operation, Soldering
Analysis Tools:	ANSYS Workbench / Mechanical / Fluent, Excel, Hand Calculations

Projects

Rocket Engine Development July 2025 – Present

- Designed a N_2O/ethanol 1.6kN liquid engine performing thermochemical analysis and nozzle optimization for 0.9 atm
- Conducted transient thermal analysis to size copper combustion chamber walls as a heat sink, calculating thermal penetration depth and temperature gradients for target burn duration of ~7 seconds.
- Performed CFD in ANSYS Fluent to extract discharge coefficients from mass flow predictions and spray cone angle
- Produced GD&T drawings for all CNC-machined components, outsourcing fabrication to third-party vendors

Active Aerodynamic Control System Jun 2024 – May 2025

- Executed parametric CFD simulations in ANSYS Fluent across deployment angles (0-100%) and Mach regimes (0.3 - 0.7), characterizing drag force profiles to support flight control algorithm development geometry and torque
- Designed a four-bar linkage mechanism for drag control, performing kinematic analysis to optimize deployment
- Developed a regression model in MATLAB from CFD, enabling fast drag lookup for the PID control system
- Validated CFD-predicted force through experimental flight testing, comparing data against simulation results
- Achieved 26% mass reduction in CNC-machined aluminum components through iterative FEA-driven design optimization while maintaining structural safety factors, allowing to stay within the team derived mass requirement.

Experience

Chief Engineer – IREC 10K Student Competition, USF Rocketry Team (SOAR) May 2025 - Present

- Responsible for technical development, systems engineering and integration of 7 subsystems in student-built rocket
- Supervised all manufacturing operations of 25+ aluminum components via CNC programming, vertical mill operations, drill press work, and tapping, ensuring dimensional accuracy, resulting in the year with most in house made parts
- Conducted system design reviews to confirm readiness, coordinating with faculty mentors and competition judges
- Established top-level mission requirements and design constraints through trade studies and competition rule analysis
- Mentored 12+ junior members in CAD/CAM, CNC machining, FEA, CFD, and systems engineering principles

Computational Aerodynamics Researcher, USF BioAero Lab – Numerical Airfoil Aerodynamics Modeling Jan 2026 – Present

- Currently developing unsteady vortex-lattice framework (UVLM) to capture time dependent lift response
- Extended a steady thin-airfoil vortex lattice MATLAB tool to model finite-thickness airfoils, modifying geometric discretization and boundary condition enforcement over different angles of attack, obtaining change in flow regime and lift coefficient.

Instrumentation Researcher, USF Corrosion Research Laboratory – Corrosion prediction in tendons May 2024 – Dec 2025

- Designed a 2-layer PCB using EasyEDA and SMD components, reducing assembly complexity and cost replication
- Developed FFT-based signal processing algorithms in Python to detect voids in post-tensioned tendons by comparing measured impedance against reconstructed baseline signatures, implementing filtering and polynomial correction
- Developed a real-time wireless data acquisition pipeline with Arduino-to-PC serial communication (UART) at 115200 baud, processing extracted sensor data in Python with automated anomaly detection and live visualization
- Designed 3D-printed enclosures and sensor mounting hardware for portable field deployment, enabling rapid prototyping iterations and tool-free assembly for bridge structure inspections

Research Lab Assistant, RANCS Research Group – Autonomous Systems Dec 2023 – Mar 2024

- Designed and built a t-slot aluminum frame to elevate a $10,000 LIDAR sensor, improving field of view and eliminating interferences with vehicle roof, enabling 100+ hours of successful autonomous vehicle testing
- Coordinated with PI and graduate students to define LIDAR mounting requirements, designing a cost-effective solution using existing lab inventory of t-slot aluminum extrusions and hardware